Assignment

(PR01)

Presented to:
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CR431 : Operating System Management Scripting with PowerShell
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1 Introduction

This semester assignment consists of building a PowerShell module dedicated to firewall log management. Firewall logs are normally long text files that are hard to read and even harder to analyze quickly. The objective of this module is to simplify this process by converting raw log data into structured and readable information.

2 Description of the Use Cases

Our module can be used for several cases. For example :

- Log analysis
- DNS resolution

3 Module Installation

To use our module, you will need to follow these steps

1. Download the provided ZIP file, which contains our PowerShell module file (FirewallLogs.psm1) and our manifest file (FirewallLogs.psd1).
2. Extract the ZIP file.
3. Open a PowerShell terminal and navigate to the folder.

```
cd C:\Users\<Your Username>\Downloads\FirewallLogs
```

4. Import our FirewallLogs module by running the following command :

```
Import-Module FirewallLogs
```

Verify the installation of our module in your system by running the following command :

```
Get-Command -ListAvailable FirewallLogs
```

4 Dependencies

In addition to the module file and the manifest file, you will need the following dependencies to use this module :

- Windows PowerShell
- Firewall log files
- Pi-hole log files

5 Commands (Functions)

5.1 Function 1: Get-FirewallLog

5.1.1 Usage

The Get-FirewallLog function reads a firewall log file, parses the contents of the log file, and displays the entries in a structured format.

5.1.2 Arguments

This function uses two arguments :

- -FirewallLogPath
  - Alias : -P
- -Full
  - Alias : -F

5.1.3 Use of arguments

The FirewallLogPath argument is used to specify the path to a firewall log file. The Full argument is optional. If you specify this argument during your command exécution, it displays all available content fields from the specified log file.

5.1.4 Format of arguments

Arguments	Format
FirewallLogPath	Type : String Format : File path
Full	Type : Switch Format : None

5.1.5 Expected result

Firewall log entries are displayed as structured PowerShell objects. If you only use the FirewallLogPath argument in the Get-FirewallLog command, you will only see the minimal fields (User, Rule name, Source IP address, Destination IP address, etc.).

5.1.6 Examples

- Example 1 : Get-FirewallLog -P "C:\Users\bryan\Downloads\FirewallLogs\fw.log"

- Example 2 : Get-FirewallLog -P "C:\Users\bryan\Downloads\FirewallLogs\fw.log" -Full

5.2 Function 2 : Find-FirewallLog

5.2.1 Usage

The Find-FirewallLog function is used to search, filter and display firewall log entries based on five specific filters :

- The source IP address
- The destination IP address
- The destination port number
- The user
- The name of the rule

5.2.2 Arguments

This function uses the following list of arguments :

- -FirewallLogPath
  - Alias : -P
- -SourceIP
  - Alias : -Src
- -DestinationIP
  - Alias : -Dst
- -DestinationPort
  - Alias : -Port
- -User
  - Alias : -User
- -RuleName
  - Alias : -Rule

5.2.3 Use of arguments

The FirewallLogPath argument tells the Find-FirewallLog function which firewall log file (with their path) to read. The other arguments are the five specified filters.

Name of the filter	The role of the filter
SourceIP	Filter the specified log file by source IP address
DestinationIP	Filter the specified log file by destination IP address
DestinationPort	Filter the specified log file by destination port number
User	Filter the specified log file by the name of the user
RuleName	Filter the specified log file by the name of the firewall rule

5.2.4 Format of arguments

Arguments	Format
FirewallLogPath	Type : String Format : File path
SourceIP	Type : String Format : IPv4 address
DestinationIP	Type : String Format : IPv4 address
DestinationPort	Type : Integer (int) Format : Port number Example : 443
User	Type : String
RuleName	Type : String

5.2.5 Expected result

Firewall log entries are returned and displayed, only if they match the filters you selected as your argument(s). The number of entry results is also displayed. If there are no matching entries, an error message will appear on your screen.

5.2.6 Examples

- Example 1 : Find-FirewallLog -P "C:\Users\user\Downloads\FirewallLogs\fw.log" -SourceIP ""

- Example 2 : Find-FirewallLog -P "C:\Users\user\Downloads\FirewallLogs\fw.log" -DestinationIP "" -DestinationPort ""

5.3 Function 3 :Resolve-FirewallDestination

5.3.1 Usage

The Resolve-FirewallDestination function is used to link destination IP addresses from firewall log files to domain names using Pi-hole log files.

5.3.2 Arguments

This function uses two arguments :

- FirewallLogPath
  - Alias : -P
- PiholeLogPath
  - Alias : -DNS

5.3.3 Use of arguments

The FirewallLogFile argument specifies the path of a firewall log file. The PiholeLogFile argument specifies the path to a Pi-hole log file.

5.3.4 Format of arguments

Arguments	Format
FirewallLogPath	Type : String Format : File path
PiholeLogPath	Type : String Format : File path

5.3.5 Expected result

The command exécution of the Resolve-FirewallDestination should output the destination IP addresses and their corresponding domain name (if there is a match). If no match is found, the appropriate domain fields will be empty.

5.3.6 Examples

- Example 1 : Resolve-FirewallDestination -P "C:\Users\user\Downloads\FirewallLogs\fw.log" -DNS "C:\Users\user\Downloads\FirewallLogs\pihole.log"
- Example 2 : Resolve-FirewallDestination -P "C:\Users\user\Downloads\FirewallLogs\fw.log.1" -DNS "C:\Users\user\Downloads\FirewallLogs\pihole.log.2"

6 Conclusion

In conclusion, our firewall log management module, FirewallLogs, makes it much easier to work with firewall logs by converting raw log data into readable log information. It can help you save you a lot of time if you need to filter and investigate network activity by analyzing log files.

7 References

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